

IMAGE ENHANCEMENT



- ▶ There exist many techniques that can enhance a digital image without spoiling it.
- ▶ The enhancement methods can broadly be divided in to the following two categories:

- Spatial Domain Methods

- Frequency Domain Methods

- ▶ Unfortunately, there is no general theory for determining what is 'good' image enhancement when it comes to human perception. If it looks good, it is good.
- ▶ However, when image enhancement techniques are used as pre-processing tools for other image processing techniques, then quantitative measures can determine which techniques are most appropriate.

FREQUENCY DOMAIN ENHANCEMENT

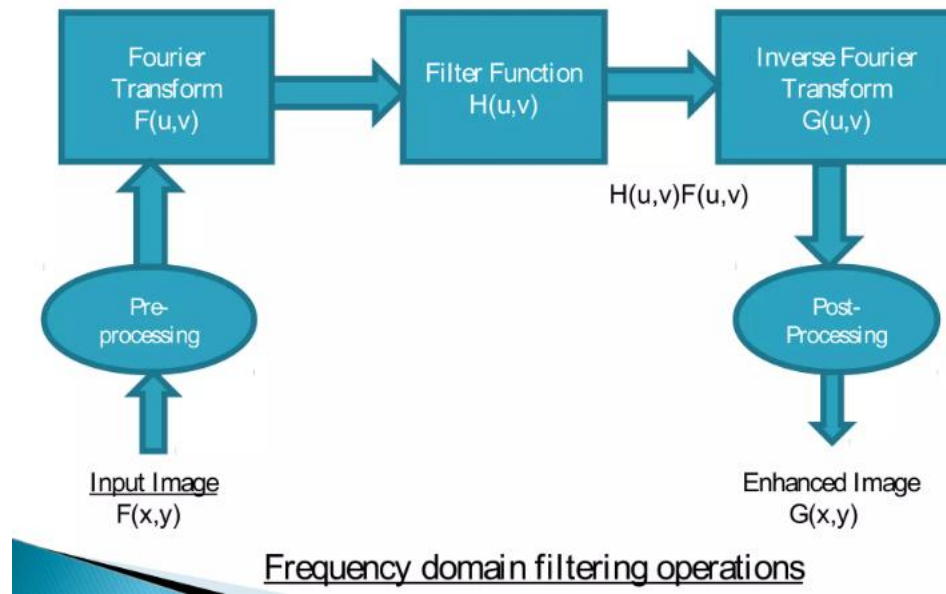


- ▶ In frequency domain methods, the image is first transferred into frequency domain.
- ▶ It means that, the Fourier Transform of the image is computed first.
- ▶ All the enhancement operations are performed on the Fourier transform of the image and then the Inverse Fourier transform is performed to get the resultant image.

- ▶ These enhancement operations are performed in order to modify the image brightness, contrast or the distribution of the grey levels.
- ▶ As a consequence the pixel value (intensities) of the output image will be modified according to the transformation function applied on the input values.
- ▶ Image enhancement is applied in every field for example medical image analysis, analysis of images from satellites etc.
- ▶ Image enhancement simply means, transforming an image F into image G using T . (Where T is the transformation function.
- ▶ The values of pixels in images F and G are denoted by r and s , respectively. As said, the pixel values r and s are related by the expression

$$s = T(r)$$

- Where T is a transformation that maps a pixel value r into a pixel value s .



- We can therefore directly design a transfer function and implement the enhancement in the frequency domain as follows.

$$G(u, v) = H(u, v)F(u, v)$$

Transfer function

Enhanced Image

Given Image

FILTERING

- The concept of filtering is easier to visualize in the frequency domain.
- Therefore, enhancement of image can be done in the frequency domain, based on its DFT.

- ▶ Filtering can be divided in two categories namely
- ▶ Low pass filtering
- ▶ High Pass filtering
- ▶ Edges and sharp transitions in gray values in an image contribute significantly to high-frequency content of its Fourier transform.
- ▶ Regions of relatively uniform gray values in an image contribute to low-frequency content of its Fourier transform.

LOW PASS FILTERING

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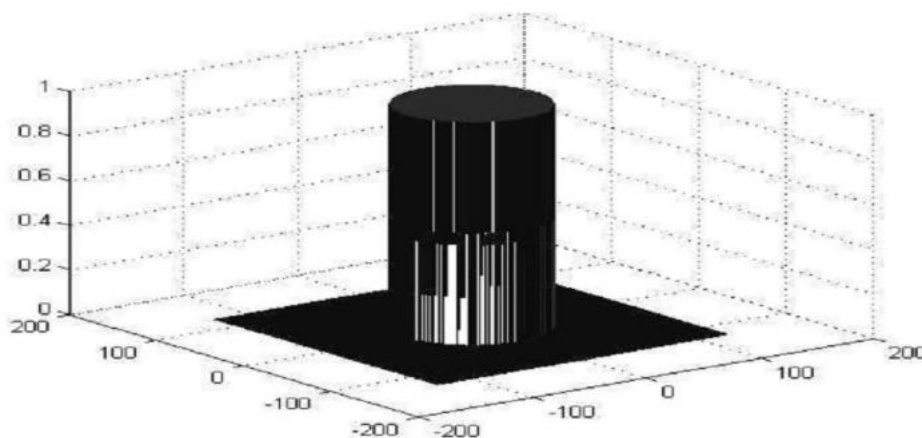
- ▶ Hence, an image can be smoothed in the Frequency domain by attenuating the high-frequency content of its Fourier transform.
- ▶ This would be a low pass filter.
- ▶ For simplicity, we will consider only those filters that are real and symmetric.

LOW PASS FILTERING

- ▶ An ideal low pass filter with cutoff frequency ' r_0 ' is given by following relation.

$$H(u, v) = \begin{cases} 1, & \text{if } \sqrt{u^2 + v^2} \leq r_0 \\ 0, & \text{if } \sqrt{u^2 + v^2} > r_0 \end{cases}$$

LOW PASS FILTERING



Ideal LPF with $r_0 = 57$

LOW PASS FILTER EXAMPLE



Original Image



LPF image, $r_0 = 57$

LOW PASS FILTER EXAMPLE



LPF image, $r_0 = 36$



LPF image, $r_0 = 26$

LOW PASS FILTERING



- ❖ The cutoff frequency of the ideal LPF determines the amount of frequency components passed by the filter.
- ❖ Smaller the value of r_0 , more the number of image components eliminated by the filter.
- ❖ In general, the value of r_0 is chosen such that most components of interest are passed through, while most components not of interest are eliminated.

BUTTERWORTH LOW PASS FILTERING

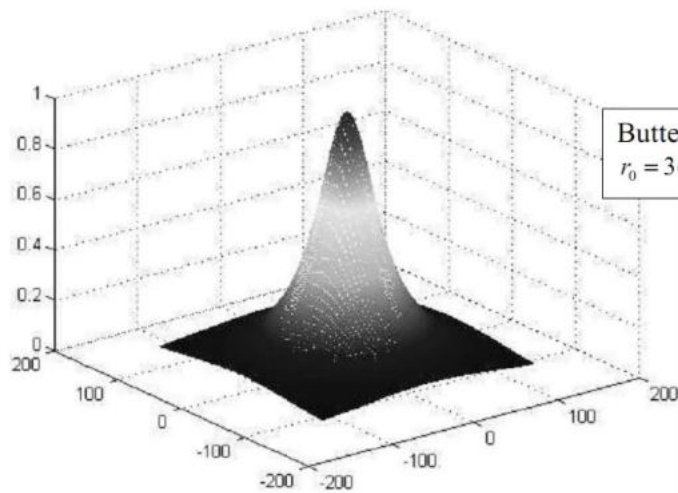


- ▶ A two-dimensional Butterworth low pass filter has transfer function:

$$H(u, v) = \frac{1}{1 + \left[\frac{\sqrt{u^2 + v^2}}{r_0} \right]^{2n}}$$

 n : filter order, r_0 : cutoff frequency

BUTTERWORTH LPF



Butterworth LPF with
 $r_0 = 36$ and $n = 1$

BUTTERWORTH LOW PASS FILTERING



- ▶ Frequency response does not have a sharp transition as in the ideal LPF.
- ▶ This is more appropriate for image smoothing than the ideal LPF, since this not introduce ringing.

BUTTERWORTH LPF EXAMPLES



Original Image



LPF image, $r_0 = 18$

BUTTERWORTH LPF EXAMPLES



LPF image, $r_0 = 13$



LPF image, $r_0 = 10$

GAUSSIAN LOW PASS FILTERING



- ▶ The form of a Gaussian low pass filter in two-dimensions is given by

$$H(u, v) = e^{-D^2(u, v) / 2\sigma^2} \quad D(u, v) = \sqrt{u^2 + v^2}$$

□ Where D is the distance from origin in frequency plane

❖ The parameter σ measures the dispersion of the Gaussian curve. Larger the value of σ , larger the cutoff frequency and milder the filtering.

HIGH PASS FILTERING



- ▶ Hence, an image can be smoothed in the Frequency domain by attenuating the low-frequency content of its Fourier transform.
- ▶ This would be a high pass filter.
- ▶ For simplicity, we will consider only those filters that are real and symmetric

HIGH PASS FILTERING

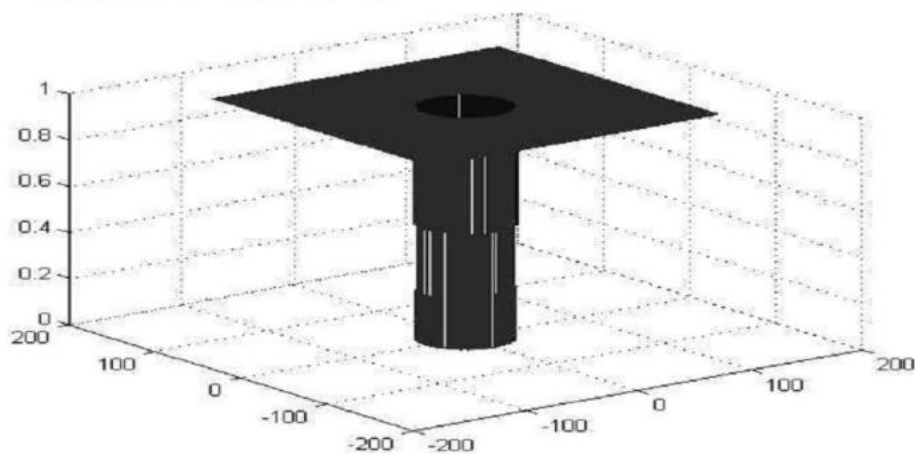
- ▶ An ideal high pass filter with cutoff frequency r_0 .

$$H(u, v) = \begin{cases} 0, & \text{if } \sqrt{u^2 + v^2} \leq r_0 \\ 1, & \text{if } \sqrt{u^2 + v^2} > r_0 \end{cases}$$

HIGH PASS FILTERING



Ideal HPF with $r_0=36$

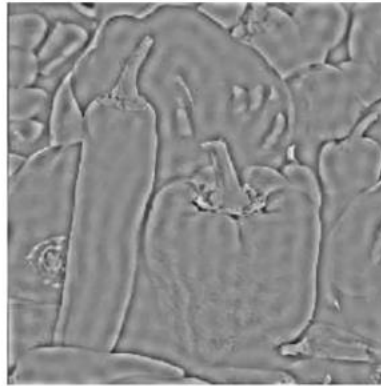


IDEAL HPF EXAMPLES

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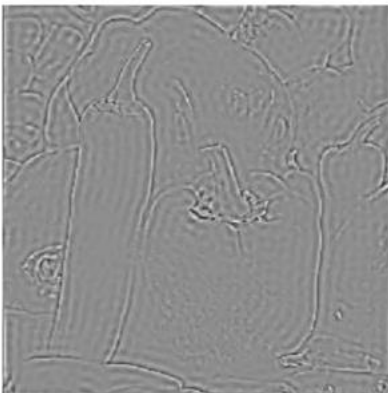
Original Image



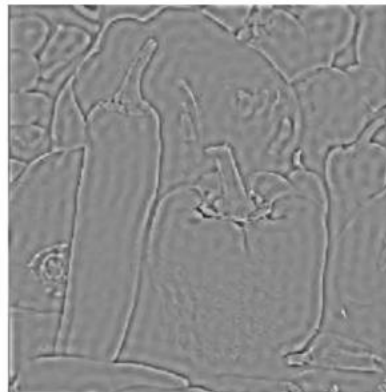
HPF image, $r_0 = 18$

IDEAL HPF EXAMPLES

W



HPF image, $r_0 = 36$



HPF image, $r_0 = 26$

BUTTERWORTH HIGH PASS FILTERING

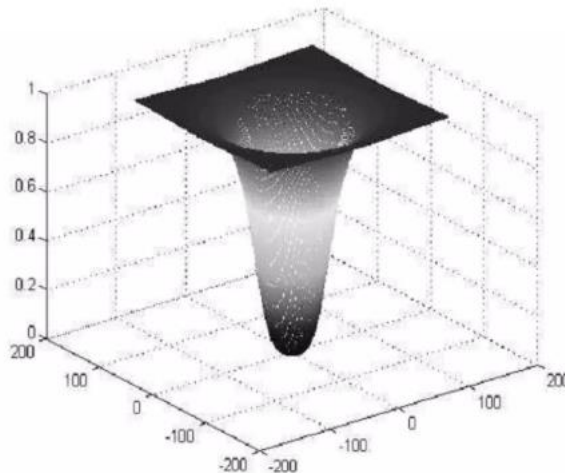
- ▶ A two-dimensional Butterworth high pass filter has transfer function:

$$H(u, v) = \frac{1}{1 + \left[\frac{r_0}{\sqrt{u^2 + v^2}} \right]^{2n}}$$

n : filter order, r_0 : cutoff frequency

BUTTERWORTH HIGH PASS FILTERING

- ❖ Butterworth HPF with cut off frequency $r_0=47$
- ❖ Order =2



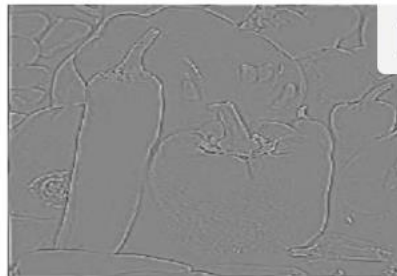
BUTTERWORTH HIGH PASS FILTERING



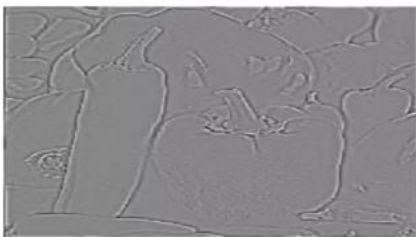
- ▶ Frequency response does not have a sharp transition as in the ideal HPF.
- ▶ This is more appropriate for image sharpening than the ideal HPF, since this not introduce ringing.
- ▶ Example of Butterworth high pass filtering is given on next slide.



Original Image



HPF image, $r_0 = 47$



HPF image, $r_0 = 36$



HPF image, $r_0 = 81$

CONCLUSION



- ▶ Image enhancement algorithms offer a wide variety of approaches for modifying images to achieve visually acceptable images.
- ▶ The choice of such techniques is a function of the specific task, image content, observer characteristics, and viewing conditions.